

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 8 (900 MHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Dr. Lena Kowalski
Reviewer:	Ana Reyes
Design Center:	Singapore RF Lab
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 8 (900 MHz) for UWB (Ultra-Wideband) Filter. The filter is designed on AlN on Si substrate with a target center frequency of 5328.8 MHz and bandwidth of 276.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 8 (900 MHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: AlN on Si
- Package: SiP (System in Package)
- Design Tool: Keysight ADS 2024
- Design Center: Singapore RF Lab

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 8 (900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5328.8 MHz
Bandwidth (3 dB):	276.9 MHz
Insertion Loss Target:	≤ 1.9 dB
Return Loss Target:	≥ 14.0 dB
Out-of-Band Rejection:	≥ 25.0 dB
Isolation (Duplexer):	≥ 30.0 dB
Q Factor:	13870
Temperature Coefficient:	-8.7 ppm/°C
Die Size:	2.7 x 2.1 mm
Wafer Size:	8-inch
Number of Resonators:	3
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 5328.8 MHz, BW 276.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15986 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.9 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 415 MHz offset >= 25.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 5.8 MHz
- Power handling shall be >= +33 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-08-30
Devices Tested: 50
Insertion Loss (meas): 1.55 dB
Return Loss (meas): 16.4 dB
Rejection (meas): 28.1 dB
Isolation (meas): 35.5 dB
Q Factor (meas): 13870
Group Delay Variation: 6.5 ns
Power Handling: +33 dBm
Die Yield: 87.1%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated excellent performance for Band 8 (900 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author:	Dr. Lena Kowalski	Date:	2024-08-30
Reviewer:	Ana Reyes	Date:	2024-08-31
Approver:	Dr. Karen Mitchell	Date:	2024-09-13

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